

Static Linking

Wednesday, October 26, 2022 5:53 PM

- Let's Create a file "MyPrint.cpp"
- `__declspec(dllexport)` to say when we make a library file or a dll we want this visibly callable!

```
#include <iostream>
using namespace std;

__declspec(dllexport) void Hello()
{
    cout << "Hello" << endl;
}
```

- Let's Create a file "Source.cpp"

```
void Hello();

int main() {
    Hello();
    return 0;
}
```

- Below command will generate *.obj file!
- We can ship this *.obj file. Client can use this *.obj file as a part of compilation process. But it will be headache (Imagine if there are 100's of files, each producing 100's of *.obj files!)

```
cl /c MyPrint.cpp
```

```
10/26/2022 06:06 PM <DIR> .
10/26/2022 06:06 PM <DIR> ..
10/26/2022 06:06 PM      112 MyPrint.cpp
10/26/2022 06:06 PM    77,612 MyPrint.obj
10/26/2022 05:59 PM      54 Source.cpp
          3 File(s)      77,778 bytes
          2 Dir(s)  778,707,484,672 bytes free
```

Static Linking

- We can create a static library using below command and by using the above generated *.obj files!

```
lib /out:MyLibrary.lib MyPrint.obj
```

- The above command will create a library file. It will take the *.obj file and pump the content into the library.
- Now client program which is "Source.cpp" can statically link this!

```

10/26/2022 06:12 PM <DIR> .
10/26/2022 06:12 PM <DIR> ..
10/26/2022 06:12 PM 111,810 MyLibrary.lib
10/26/2022 06:11 PM 114 MyPrint.cpp
10/26/2022 06:06 PM 77,612 MyPrint.obj
10/26/2022 05:59 PM 54 Source.cpp
4 File(s) 189,590 bytes
2 Dir(s) 778,686,095,360 bytes free

```

cl Source.cpp /link MyLibrary.lib

- The above command will create "Source.exe" and client can run this.

```

10/26/2022 06:16 PM <DIR> .
10/26/2022 06:16 PM <DIR> ..
10/26/2022 06:12 PM 111,810 MyLibrary.lib
10/26/2022 06:11 PM 114 MyPrint.cpp
10/26/2022 06:06 PM 77,612 MyPrint.obj
10/26/2022 05:59 PM 54 Source.cpp
10/26/2022 06:16 PM 178,176 Source.exe
10/26/2022 06:16 PM 707 Source.exp
10/26/2022 06:16 PM 1,708 Source.lib
10/26/2022 06:16 PM 677 Source.obj
8 File(s) 370,858 bytes
2 Dir(s) 778,683,998,208 bytes free

C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking>Source.exe
Hello

```

Size is more

- Now even if we erase the MyLibrary.lib file and again run the Source.exe, it will work. This is because the compiler copied and pasted the machine instructions into the executable.

```

C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking>erase MyLibrary.lib

C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking>dir
Volume in drive C is Local Disk
Volume Serial Number is 426D-16DE

Directory of C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking

10/26/2022 06:20 PM <DIR> .
10/26/2022 06:20 PM <DIR> ..
10/26/2022 06:11 PM 114 MyPrint.cpp
10/26/2022 06:06 PM 77,612 MyPrint.obj
10/26/2022 05:59 PM 54 Source.cpp
10/26/2022 06:16 PM 178,176 Source.exe
10/26/2022 06:16 PM 707 Source.exp
10/26/2022 06:16 PM 1,708 Source.lib
10/26/2022 06:16 PM 677 Source.obj
7 File(s) 259,048 bytes
2 Dir(s) 778,678,632,448 bytes free

C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking>Source.exe
Hello

```